

Question Paper Preview

Notations :

- 1.Options shown in **green** color and with  icon are correct.
- 2.Options shown in **red** color and with  icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Computer System Design

Section Id :	640653124702
Number of Questions :	21
Section Marks :	50
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531863459 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : COMPUTER SYSTEM DESIGN (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536036616.  YES

6406536036617.  NO

Question Number : 2 Question Id : 6406531863460 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A multiplexer is a combinational circuit that selects one input from multiple inputs and forwards it to a single output.

Options :

6406536036618. ✓ True

6406536036619. ✗ False

Question Number : 3 Question Id : 6406531863461 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A JK flip-flop cannot store information without a continuous clock input.

Options :

6406536036620. ✗ True

6406536036621. ✓ False

Question Number : 4 Question Id : 6406531863462 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In a 4-bit system, the two's complement of a binary number 1010 can be represented as -6.

Options :

6406536036622. ✓ True

6406536036623. ✗ False

Question Number : 5 Question Id : 6406531863463 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A decoder performs the inverse operation of a multiplexer.

Options :

6406536036624. ✓ True

6406536036625. ✗ False

Question Number : 6 Question Id : 6406531863464 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

According to Moore's Law, the size of transistors on an integrated circuit doubles approximately every two years.

Options :

6406536036626. ✖ True

6406536036627. ✔ False

Question Number : 7 Question Id : 6406531863465 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Using 2's complement method, evaluate $A - B$ where $A = 180$ and $B = 50$. Give the answer in 8-bit binary format.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

01100010

Question Number : 8 Question Id : 6406531863466 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The circuit diagram requires the output

$$f = (\overline{x} + \overline{y})zw$$

The gates GT1 and GT2 should be.

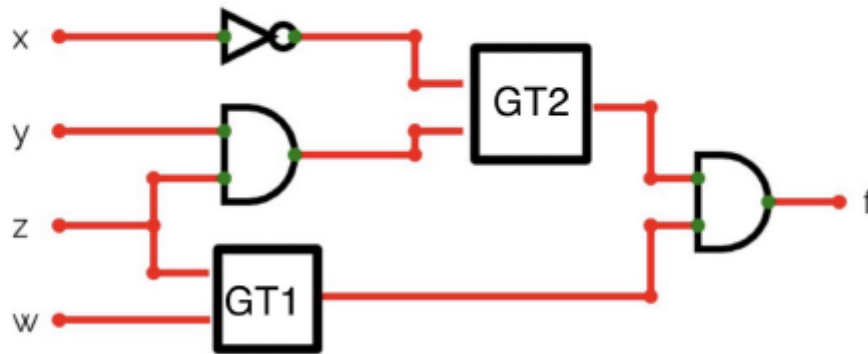


Figure 1: Circuit Diagram

Options :

6406536036629. ✖ OR,NAND

6406536036630. ✖ OR,NOR

6406536036631. ✖ AND,NOR

6406536036632. ✔ AND,NAND

6406536036633. ✖ None of these

Question Number : 9 Question Id : 6406531863467 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The solutions of the quadratic equation

$$x^2 - 11_b x + 21_b = 0,$$

are $x = 3$ and $x = 5$. What is the base b of the numbers?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

Question Number : 10 Question Id : 6406531863468 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

What is the minimum number of 2-input NAND gates required to implement the circuit for minimized SOP expression of the function.

$$f(x, y, z) = \sum m(0, 2, 5, 7)$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 11 Question Id : 6406531863469 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Given that NOT, 2-input AND/OR gates have delays of 1 ns, what will be the maximum delay of a 2x1 multiplexer circuit?

Note: if the answer is 1 ns, write 1.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

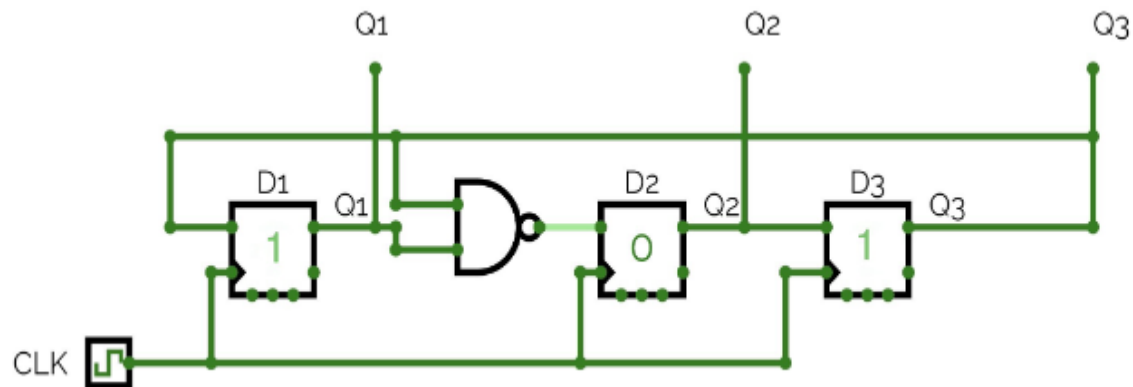
3

Question Number : 12 Question Id : 6406531863470 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider the circuit consisting of three D flipflops and one NAND gate as shown below. The initial states are $Q_1=1$, $Q_2=0$, and $Q_3=1$. What is the output after (Q_1, Q_2, Q_3) after 5 clock pulse?



Note: If answer is $Q_1=0$, $Q_2=0$, $Q_3=0$, then write 000

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

101

Question Number : 13 Question Id : 6406531863471 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

A processor uses the following control word format:

Opcode	Src1	Src2	Dest
10	1011	1110	0001

How many registers can be addressed in the register bank?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

Question Number : 14 Question Id : 6406531863472 Question Type : SA
Correct Marks : 3

Question Label : Short Answer Question

Consider a common bus architecture containing memory and three registers: A, B, and C.
The control word format is shown below.

Memory		Register A		Register B		Register C	
Read	Write	Input	Output	Input	Output	Input	Output

What would be the control word if you intend to perform a register transfer from register B to register C?
Note: Provide your answer in hexadecimal format (e.g., 0x3A).

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

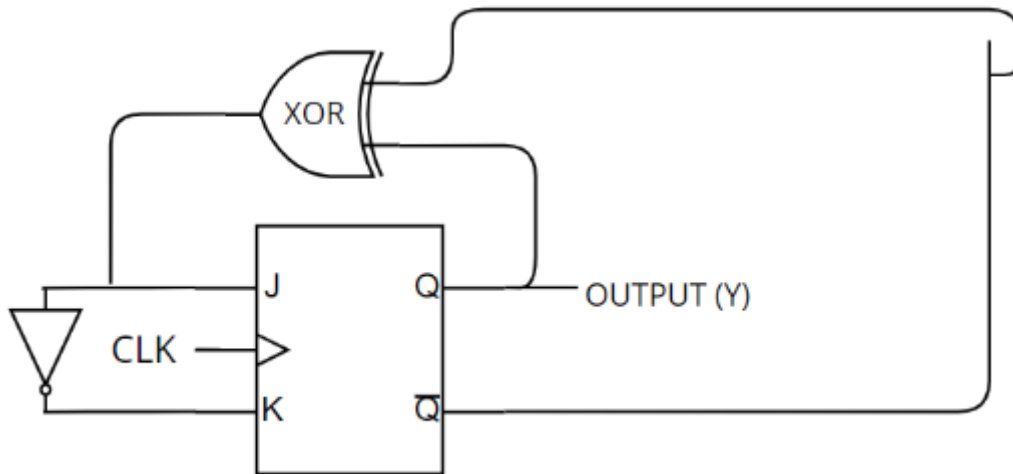
0x06

Question Number : 15 Question Id : 6406531863473 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider the circuit diagram as shown below. Assuming it was initially cleared to 0, state the output(Y) for the next 4 clock pulses.



Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1111

Question Number : 16 Question Id : 6406531863474 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Choose the correct truth table according to the given Boolean expression:

$$F(x, y, z) = \overline{x}y\overline{z} + \overline{x}y + x\overline{z}$$

Options :

x	y	z	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

6406536036641. ✖

6406536036642. ✖ None of these

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

6406536036643. ✔

6406536036644. ✖

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Question Number : 17 Question Id : 6406531863475 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

A modern RISC processor contains 32 registers in its general-purpose register file. The Arithmetic Logic Unit (ALU) of the processor is capable of executing 8 distinct operations, which include standard arithmetic and logical functions.

What is the *minimum* number of bits required by a ~~three-address~~ instruction to perform a bitwise **OR** operation between two source operands located in the register file, and store the computational result back into a designated destination register?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

18

Question Number : 18 Question Id : 6406531863476 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose we want to execute a data movement instruction, where we move the data present in register R_{20} to register R_{15} .

Instruction: $\text{MOV } R_d, R_r$

R_d : Destination register, R_r : Source register

Operands: $0 \leq d \leq 31, 0 \leq r \leq 31$

Instruction Format (16-bit):

0010	11rd	drdr	rrdd
------	------	------	------

What will be the hexadecimal machine code for the instruction? Note: Enter the value in hexadecimal format (e.g. 0x24E8).

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

0x2EB3

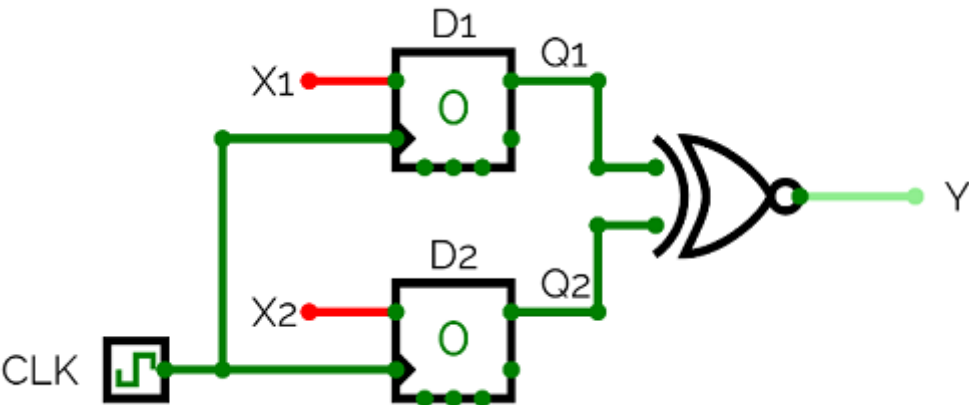
Question Number : 19 Question Id : 6406531863477 Question Type : MCQ

Correct Marks : 3

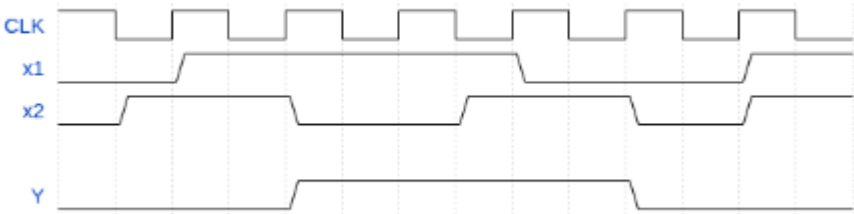
Question Label : Multiple Choice Question

In the circuit shown below, D1 and D2 are positive triggered D flipflops, x1 and x2 are inputs of D1 and D2 flipflop.

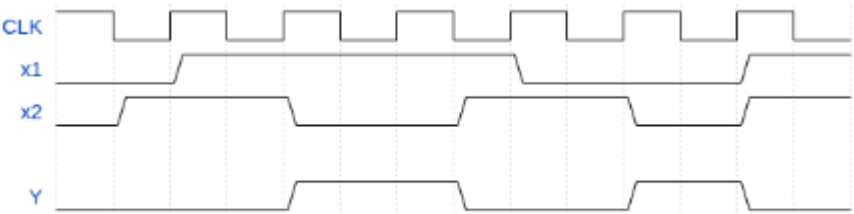
Choose the correct timing diagram of the output Y.



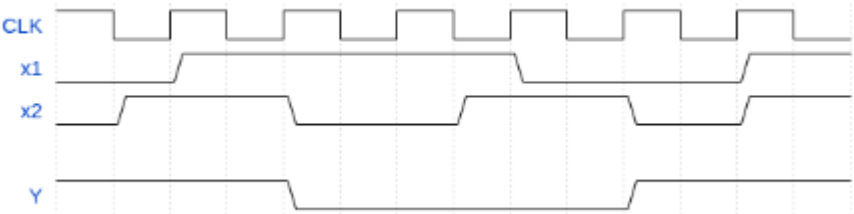
Options :



6406536036647. ✖

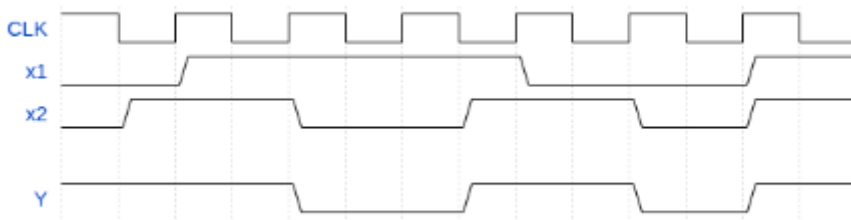


6406536036648. ✖



6406536036649. ✔

6406536036650. ✖



6406536036651. ✖ None of these

Question Number : 20 Question Id : 6406531863478 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider the function given below:

$$F(A, B, C, D) = \sum(1, 2, 3, 5, 6, 7, 9, 10, 11)$$

If we want to implement the minimized expression using only 2-input NAND gates, what is the minimum number of NAND gates required?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

Question Number : 21 Question Id : 6406531863479 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a gated D-Latch constructed using NAND gates. The initial input (at 0 ns) is D = 1, and the write enable is high. The write enable toggles between high and low every 5 ns. The input D changes from 0 to 1 and from 1 to 0 every 6 seconds. What will be the output Q after 22 ns?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0